

Human neural-circuit model — Pass 1

A note on what this is before the deliverables. The **backbone** below is canonical systems/affective/developmental neuroscience — textbook-stable, and I've written it from established knowledge. I ran targeted searches only where recent literature actually moves the picture: the contested framings, the human-vs-animal splits, and the developmental-timing numbers (those citations appear inline). Everything is tagged **E / Em / H** (established / emerging / hypothesised) and labelled for human vs animal evidence.

Scope of this pass (fully covered): defensive/threat · reward/motivation · stress · affiliation/attachment · executive control · arousal/regulatory — plus interoception, episodic memory, and the CSTC loop scaffolding, because the network and developmental layers need them. **Deferred to later passes** (flagged at the end): attention (as its own registry block), working-memory decomposition, social cognition/mentalizing, language, sensory (visual/auditory/somatosensory), and full motor control.

The co-activation matrix and developmental-parameter table are generated files (previewed below, attached at the end) so they load cleanly and stay internally consistent.

0 · Schema (defined before populating)

Registry record (Deliverable 1) — fields per circuit

Field	Content
ID · Name	Stable handle + label (e.g. THREAT-01 · Amygdala acute-threat)
Nodes → projections	Key nuclei/regions and the directed connections between them
Transmitters	Principal neuromodulators/transmitters
Function	Behaviour/process produced or supported
RDoC · Panksepp	RDoC domain·construct, and primary-process system where applicable
Dev & plasticity	Brief onset/maturation + plasticity type; full parameters in §4 file
Evidence · Conf	Human/animal basis + confidence tag

Co-activation matrix (Deliverable 2) — layout. Rows = states/events; columns = circuit IDs; cells = signed ordinal weight. Legend (a **qualitative expert-consensus scaffold — not measured effect sizes or coupling coefficients**):

+2 drive (primary) · +1 modulate (engaged/supporting) · 0/blank not-consistently-engaged-or-unknown · -1 inhibit (mild/suppressed) · -2 inhibit (strong/offline). Per-edge confidence lives in the long/JSON forms.

Network map (Deliverable 3): network → core nodes → constituent circuit IDs → role → contested notes.

Developmental layer (Deliverable 4): per-circuit parameter row (onset · milestone · functional_maturation · pruning_window · myelination · sensitive_period_gates · plasticity_type · plasticity_coeff · confidence · evidence_base) plus rule notes.

RDoC is used as the structuring spine. Its current matrix has **six domains — Negative Valence, Positive Valence, Cognitive, Social Processes, Arousal/Regulatory, Sensorimotor — with constructs presented explicitly as evolving exemplars rather than fixed categories**, the matrix having been built to change as research accrues, with the Sensorimotor domain added as the sixth. [SpringerNational Institute of Mental Health](#)

1 · Circuit registry

A. Defensive / Threat — RDoC Negative Valence

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
THREAT-01 Amygdala acute-threat	Sensory thalamus & cortex → lateral amygdala (LA) → basal (BA) → central (CeA); CeA → hypothalamus, PAG	Glu/GABA microcircuit; modulated by NA, DA, 5-HT; intra-amygdala peptides (CRF)	Rapid detection of/response to imminent threat; Pavlovian fear conditioning; autonomic + behavioural mobilisation	NegVal · Acute Threat ("Fear") · Panksepp FEAR	Functional in infancy; fear-learning re-tuned by vmPFC into adolescence; EE+ED	Human+animal · E (LeDoux reframes CeA output as "defensive survival circuit"; conscious <i>fear</i> is a separate cortical construction)
THREAT-02 BNST sustained-threat	CeA → bed nucleus of stria terminalis (BNST; "extended amygdala")	CRF-dominant; Glu/GABA; A; NA	Sustained/uncertain/temporally-distant threat; anxiety-like apprehension (vs phasic fear)	NegVal · Potential Threat ("Anxiety")	Matures later than CeA; childhood adversity biases	Mostly animal (Davis) · E (animal) / Em (human)

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
	) → hypothalamus, PAG, LC				toward this mode	
THREAT-03 PAG defensive output	Dorsal/lateral PAG (flight/freeze), ventrolateral PAG (quiescence); descending to medulla/spinal	Glu/GABA; opioids; NA	Selects & executes innate defensive behaviour (freeze/flight/fight) and stress-analgesia	NegVal · Acute/Potential Threat · FEAR/RAGE output	Innate, functional near birth; low plasticity	Human+animal · E
THREAT-04 vmPFC/IL extinction & regulation	Ventromedial PFC / infralimbic -analogue → amygdala intercalated cells; hippocampal context input	Glu → GABAergic ITCs; DA/NA modulation	Extinction learning, safety signalling, top-down regulation of threat	NegVal (regulation) · Cognitive Control overlap	Weak in childhood; PFC-paced maturation into mid-20s; ED	Human+animal · Em (human causal timing)
THREAT-05 Defensive / affective aggression	Medial amygdala → hypothalamus (attack area) → dorsal PAG	Glu/GABA; substance P; NA; testosterone-sensitive	Affective/defensive aggression; irritable "hot" aggression	NegVal (Frustrative Nonreward overlap) · Panksepp RAGE	Innate substrate; social experience shapes threshold	Mostly animal · Em

B. Reward / Motivation — RDoC Positive Valence · Panksepp SEEKING

ID · Name	Nodes → projection	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
REW-01 Mesolimbic DA / RPE	Ventral tegmental area (VTA) → nucleus accumbens (NAc); GABA/Glu afferents; lateral hypothalamus, PPTg	Dopamine (phasic); Glu/GABA	Reward-prediction-error teaching signal; reinforcement learning; approach	PosVal · Reward Learning / Responsiveness · SEEKING	DA very active early; striatal DA & receptor density peak ~adolescence then prune	Human+animal · E (Schultz RPE; ongoing debate on DA also coding salience/novelty)
REW-02 Mesocortical DA	VTA → mPFC, OFC, dACC	Dopamine (tonic, D1/D2)	Value/effort/cost integration; DA modulation of PFC working memory (inverted-U)	PosVal · Reward Valuation; Cognitive overlap	Matures with PFC into mid-20s; ED	Human+animal · E/Em
REW-03 NAc incentive-salience	NAc shell/core; ventral pallidum; hedonic "hotspots"	DA ("wanting"); μ-opioid / endocannabinoid ("liking"); GABA	Dissociable incentive-salience ("wanting") vs hedonic impact ("liking"); motivational vigour	PosVal · Reward Responsiveness (Anticipation/Initial Response/Satiation) · SEEKING	Hedonics present in infancy; wanting-system tunable	Human+animal · E (Berridge wanting≠liking)
REW-04 OFC/vmPFC value	Orbitofrontal & ventromedial PFC ↔ amygdala, NAc, mediodorsal thalamus	Glu; DA/5-HT modulation	Represents subjective/economic value & outcome expectancy; goal-directed valuation	PosVal · Reward Valuation	Protracted; late-adolescence to mid-20s; ED	Human+animal · E

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
REW-05	Sensorimotor cortex → Dorsolateral striatal habit	dorsolateral DA (consolidating); Glu/GABA → GPi/SNr → thalamus	Stimulus—response habit; automatization as goal-directed control wanes	Sensorimotor · Habit (formation cross-listed to PosVal)	Strengthens with repetition lifelong; no fixed window; ED	Human+animal · E

C. Stress — RDoC Arousal/Regulatory + Negative Valence

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
STR-01 HPA neuroendocrine axis	Hypothalamic PVN (CRH) → anterior pituitary (ACTH) → adrenal cortex (cortisol); feedback via hippocampus, mPFC, amygdala	CRH, ACTH, glucocorticoids; AVP co-secreta	Slow (minutes—hours) systemic stress response; energy mobilisation; glucocorticoid negative feedback; allostatic load when chronic	Arousal/Regulatory; NegVal (Sustained Threat)	Axis formed prenatally; stress-hyporesponsive period in infancy; set-points programmed early	Human+animal · E (McEwen allostatic load)
STR-02 CRF extended-amygdala	CeA/BNST CRF neurons → LC, PVN, PAG	CRF; NA cross-talk	Fast central stress/anxiety drive; couples threat appraisal	NegVal · Sustained Threat	Upregulated by early adversity; ED	Mostly animal · Em

ID · Name	Nodes → projections	Transmitter s	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
			to arousal & HPA			

(LC-NA
acute stress
= **AROU-01**,
cross-listed
in F)

D. Affiliation / Attachment — RDoC Social · Panksepp CARE, PANIC/GRIEF, LUST, PLAY

ID · Name	Nodes → projections	Transmitter s	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
AFF-01 Oxytocin/vasopressin bonding	PVN/SON (OT, AVP) → NAc, VTA, amygdala, BNST, lateral septum	Oxytocin, vasopressin ; gates DA reward	Pair/social bonding; social salience; amygdala threat-damping	Social · Affiliation & Attachment · (substrate for CARE/LUST)	Receptor patterning set early; caregiving may organise it	Animal-dominant (prairie-vole pair-bonding) · E (animal). Human intranasal-OT effects are weak and inconsistent — a large fraction of tested outcomes show no main effect and most show situational/individual moderators , ScienceDirect and the

ID · Name	Nodes → projections	Transmitter s	Function	RDoC · Pankse pp	Dev & plasticity	Evidence · Conf
						only meta-analysis on intranasal OT and trust found no effect, with estimates that a large share of reported interaction effects may be false positives. biorxiv Tag human translation Em/contexted.
AFF-02 CARE / parental-nurturance	Medial preoptic area (MPOA) → VTA-NAc, BNST, PAG, ACC	Oxytocin, prolactin, dopamine, est/prog	Parental motivation & caregiving; nurturant approach	Social · Affiliation & Attachment · CARE	Hormonally gated (pregnancy/parenting); experience-tuned	Mostly animal · Em
AFF-03 PANIC/GRIEF separation-distress	Dorsal PAG, anterior cingulate, BNST, dorsomedial thalamus, ventral septum	Endogenous opioids (↓ distress), oxytocin, prolactin (↓); CRF (↑)	Separation-distress ("protest") vocalisation & affect; substrate of social loss/grief	Social; NegVal (Loss) · PANIC/GRIEF	Functional early (infant cry); disruption sensitises it	Mostly animal · Em (Panksepp)
AFF-04 Social-reward	VTA → NAc for social stimuli	Dopamine, opioids, oxytocin	Makes social contact rewarding;	Social; PosVal	As REW-01	Human+animal · E/Em

ID · Name	Nodes → projections	Transmitter s	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
	(overlaps REW-01/03)		reinforces affiliation			
AFF-05 LUST / sexual-motivation	MPOA, ventromedial hypothalamus, BNST, medial amygdala	Gonadal steroids (T, E); oxytocin/vasopressin; DA	Sexual motivation & appetitive/consolidatory behaviour	Social; PosVal · LUST	Puberty-gated (steroid surge); organisational + activational	Mostly animal (human sex-specific) · Em
AFF-06 PLAY / rough-and-tumble	PAG, parafascicular/dorsomedial thalamus, somatosensory	Opioids, endocannabinoids ; DA	Social play; social-skill learning	Social · PLAY	Peaks juvenile; experience-dependent	Mostly animal · Em/H in humans

E. Executive Control — RDoC Cognitive

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
EXEC-01 dlPFC– parietal control / WM	Dorsolateral PFC ↔ posterior parietal (IPS); core of frontoparietal net	Glu; DA (D1 inverted-U), NA (α2A), ACh	Working-memory maintenance/manipulation; top-down goal control; core executive function	Cognitive · Cognitive Control / Working Memory	Among last cortex to mature; protracted into mid-20s (see §4); ED	Human · E (persistent -activity vs activity-silent WM debated)
EXEC-02 dACC conflict/error monitoring	Dorsal anterior/mid-cingulate ↔ dlPFC, pre-SMA, LC	Glu; DA; NA	Conflict & error detection (ERN), performance monitoring, specifying expected value of control	Cognitive · Cognitive Control	Matures late-adolescence–20s; ED	Human · E (Botvinick conflict; Shenhav/Cohen EVC)

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
	Right IFG + pre-					
EXEC-03	SMA → Frontostriatal response-inhibition (hyperdirect) → GPi	Glu → STN; GABA; NA	Rapid action cancellation/stopping; proactive & reactive inhibition	Cognitive; Sensorimotor (inhibition/termination)	Stopping improves to late adolescence; ED	Human+animal · E (Aron)
EXEC-04	Lateral Frontoparietal flexible-control hub	PFC + IPS + dACC (multiple-demand system) Glu; DA/NA	Rapidly instantiated task-sets; adaptive coordination; connects to DMN & DAN	Cognitive · Cognitive Control	Protracted; ED	Human · E/Em (Duncan MD system; Cole flexible hub)

F. Arousal / Regulatory — RDoC Arousal/Regulatory

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
AROU-01	LC-noradrenaline → near-global cortical/subcortical NA	Noradrenaline	Global arousal, vigilance, phasic gain/"network reset", stress; tonic- vs-phasic modes	Arousal/Regulatory · Arousal	Functional early; low-moderate plasticity	Human+animal · E (Aston-Jones/Cohen)
AROU-02	Dorsal/median raphe → widespread cortex, limbic	Serotonin	Mood, patience/impulse, aversive processing, sleep-wake modulation	Arousal/Regulatory; NegVal modulation	Early; ED	Human+animal · E

ID · Name	Nodes → projections	Transmitt ers	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
AROU-03 Basal- forebrain cholinergic	Nucleus basalis/medial septum → cortex, hippocampus	Acetylcho line	Cortical activation, attentional gain, encoding/pla sticity gating	Arousal/Regu latory; Cognitive·Att ention	Early; supports plasticity	Human+a nimal · E
AROU-04 Tuberomam millary histaminergi c	TMN (posterior hypothalamus) → cortex	Histamine	Wakefulness maintenance	Arousal/Regu latory	Early	Human+a nimal · E
AROU-05 Orexin/hypo cretin	Lateral hypothalamus orexin → LC/TMN/raphe/ VTA/BF	Orexin (hypocret in); glutamate	Stabilises wakefulness; "finger" on the sleep- wake flip- flop; loss → narcolepsy	Arousal/Regu latory · Sleep- Wakefulness	Matures across infancy w/ consolid ated sleep	Human+a nimal · E (Saper flip-flop)
AROU-06 DA wake- promoting	VTA / ventral PAG DA → forebrain	Dopamine	Wake drive, motivational arousal	Arousal/Regu latory	Early	Human+a nimal · E/Em
AROU-07 VLPO sleep- switch	Ventrolateral preoptic (GABA/galanin) → inhibits all arousal nuclei	GABA, galanin	Sleep- promoting arm of the mutually- inhibitory flip-flop switch	Arousal/Regu latory · Sleep- Wakefulness	Infancy consolid ation	Human+a nimal · E
AROU-08 SCN circadian pacemaker	Suprachiasmatic nucleus ← retinohypothala mic tract; → hypothalamus, pineal (melatonin)	VIP, AVP; melatonin output	Master ~24 h clock; entrains sleep/wake, cortisol, temperature	Arousal/Regu latory · Circadian Rhythms	Entrained postnatal ly	Human+a nimal · E

ID · Name	Nodes → projections	Transmitt ers	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
AROU-09 Thalamo- reticular ARAS gating	Brainstem reticular formation → thalamus; thalamic reticular nucleus	Glu/GABA ; ACh	Ascending arousal relay; thalamic gating of cortical state; consciousnes s level	Arousal/Regu latory	Early	Human+anim al · E

G. Interoception — feeds the Salience network

ID · Name	Nodes → projectio ns	Transmi tters	Function	RDoC · Panksepp	Dev & plasticit y	Evidence · Conf
INT-01 Insular interocep tive	Lamina-I spinothal amic → VMpo thalamus → posterior insula (primary) → mid → anterior insula (integrati on)	Glu; modulat ed by NA/5- HT/pept ides	Representation of bodily physiological condition; substrate of feelings & the material self	Arousal/Regu latory; NegVal/PosV al (bodily affect)	Posteri or early, agranul ar anterior late; EE+ED	Human+anim al · E (Craig homeostatic- afferent). Newer predictive/all ostatic framing (interoceptiv e signals partly reflect top-down limbic <i>predictions</i> , corrected by ascending afference) is Em — the Embodied Predictive Interoception Coding model casts agranular visceromotor cortices as

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
						issuing allostatic predictions rather than perception simply following sensation. PubMed CentralNature
INT-02 Salience-detection	Anterior insula + dorsal ACC (fronto-insular) ↔ amygdala, striatum, thalamus	Glu; DA/NA; von-Economo neurons	Detects behaviourally salient events (interoceptive/emotional/cognitive); proposed switchboard between internal & external modes	Arousal/Regulatory; Cognitive	Anterior insula/dACC mature late; ED	Human · Em (switching role influential but contested — see §3)

H. Episodic Memory — RDoC Cognitive · Declarative Memory

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
MEM-01 Hippocampal episodic	Entorhinal cortex → dentate gyrus → CA3 → CA1 → subiculum → EC (trisynaptic loop); + EC ↔ neocortex	Glu (NMDA-dependent LTP/LTD); ACh, DA (novelty)	Rapid encoding of episodes; pattern separation completion (DG) & (CA3); spatial/relational binding	Cognitive · Declarative Memory	DG neurogenesis is postnatal; maturation ~2–4 y gates offset of infantile amnesia; EE+ED	Human+animal · E

ID · Name	Nodes → projections	Transmitters	Function	RDoC · Panksepp	Dev & plasticity	Evidence · Conf
MEM-02 Papez / MTL- diencephalic	Hippocampus → fornix → mammillary bodies → anterior thalamus → cingulate → EC	Glu; ACh	Diencephalic limb of episodic memory; lesion → amnesia (e.g. Korsakoff)	Cognitive · Declarative Memory	Matures with hippocampus	Human+animal · E
MEM-03 Hippocampal-cortical consolidation	Hippocampus ↔ neocortex (esp. mPFC); sleep replay	Glu; sleep- dependent oscillations	Systems consolidation: gradual transfer to cortex; schema integration	Cognitive · Declarative Memory	Lifelong; sleep-gated	Human+animal · E (standard vs multiple- trace/CLS debate)

I. CSTC scaffolding — cortico-striato-thalamo-cortical parallel loops (framework)

The recurring loop motif (Alexander–DeLong–Strick): cortex → striatum → pallidum/SNr → thalamus → cortex, with a **direct** (facilitatory), **indirect** (suppressive, via GPe/STN) and **hyperdirect** (cortex→STN, fast stop) pathway per loop, all DA-modulated.

ID · Name	Cortical origin → striatal target	Function	Maps to	Conf
CSTC-MOTOR Motor loop	M1/SMA → putamen	Action selection/initiation, scaling of movement	Sensorimotor · Motor Actions	E
CSTC-OCULO Oculomotor loop	FEF/SEF → caudate (body)	Saccade selection	Sensorimotor; Attention	E
CSTC-ASSOC Associative loop	dIPFC → dorsal caudate	Executive/goal-directed cognition; working- memory gating	Cognitive	E
CSTC-ORBITO Orbitofrontal loop	OFC → ventromedial caudate	Value-based & flexible behaviour	PosVal	E

ID · Name	Cortical origin → striatal target	Function	Maps to	Conf
	ACC + vmPFC → ventral			
CSTC-LIMBIC	striatum/NAc → ventral	Motivation, reward-	PosVal; Social	E
Limbic/ACC loop	pallidum → MD	guided action, affect		
	thalamus			

Registry gaps flagged (not fabricated): attention circuits (dorsal/ventral orienting) are represented only as network nodes below, not as full registry rows — deferred. Working memory is folded into EXEC-01 rather than decomposed. Social-cognition/mentalizing (temporo-parietal junction, dmPFC, precuneus) is distinct from affiliation and is **not** yet registered. Sensory hierarchies (V1→..., A1→..., S1→...) and language circuits are deferred. Several affiliation/Panksepp anatomies (LUST, PLAY, PANIC/GRIEF precise human nodes) rest on animal work; treat human node-level detail as uncertain.

2 · State → circuit co-activation matrix

Readable summary — the strongest, best-documented patterns:

- **Acute threat** drives amygdala (THREAT-01) + PAG (THREAT-03) + LC-noradrenaline (AROU-01) together, engages HPA/CRF, and **suppresses the default-mode network** and (under high arousal) deliberative dlPFC control — the "reflexive takeover." Well established.
- **Reward anticipation vs receipt** dissociates cleanly: anticipation is dopaminergic "wanting" (REW-01/03) with attentional capture; consummatory pleasure shifts weight to NAc opioid hotspots + OFC outcome value (REW-03/04); **reward omission** drives a below-baseline DA dip plus dACC.
- **Task-focused attention** = dorsal-attention (DAN-01) + frontoparietal control (EXEC-01) up, **DMN down (anticorrelation)** — one of the most replicated findings in the field.
- **Rest/mind-wandering** is the mirror image: DMN + hippocampal engagement up, DAN/EXEC down.
- **Error/conflict** reliably drives dACC (EXEC-02) with a phasic LC-noradrenaline error response; **response inhibition** loads the right-IFG→pre-SMA→STN hyperdirect stop pathway (EXEC-03).
- **Goal-directed vs habitual action** shifts weight from OFC/vmPFC value + dlPFC (REW-04/EXEC-01) to dorsolateral-striatal S-R (REW-05) as behaviour automatises.
- **Weaker / more contested rows** (tagged Em throughout): affiliation and separation-distress (animal-dominant substrates; human oxytocin translation weak), and

empathy/social-rejection edges that lean on the anterior-insula/dACC "social pain" overlap — real signal, genuine dispute about interpretation.

Wide matrix (states × circuits; blanks = 0). Full file attached; preview of the first rows:

csv

state,THREAT-01,THREAT-02,THREAT-03,THREAT-04,REW-01,REW-02,REW-03,REW-04,REW-05,STR-01,STR-02,AFF-01,AFF-02,AFF-03,EXEC-01,EXEC-02,EXEC-03,AROU-01,INT-01,INT-02,MEM-01,DMN-01,DAN-01

acute_threat,2,,2,-1,,,,,1,1,,,,-1,,2,,1,1,-2,1

potential_sustained_threat,1,2,1,,,,,1,2,,,,1,,1,1,-1,1

reward_anticipation,,,,,2,1,2,1,,,,,,1,1,-1,1

reward_receipt,,,,,1,,2,2,,,,,,1,,,

reward_omission_frustrative_nonreward,1,,,,-2,,1,,1,,,,1,1,,,

social_bonding_affiliation,-1,,,,1,1,-1,,2,2,,,,,1,,,

social_rejection_separation,1,,1,-1,,,,,1,1,,2,,,,,2,1,

acute_psychosocial_stress,1,,,-1,,,,,2,2,,,,-1,,2,1,1,-1,

focused_external_attention,,,,,,2,1,1,1,-2,2

rest_mind_wandering,,,,,,,-1,,,,-1,1,2,-1

The **long/tidy form** (state, circuit_id, role, weight, confidence) is the richer artefact — 140 edges, each with a per-edge confidence tag (84 established, 56 emerging) — and it pivots straight back to the wide shape. JSON carries the same content with legends embedded.

These weights are a modelling scaffold, not empirical coupling coefficients — treat them as priors to be replaced by data.

3 • Circuit → network aggregation map

Network	Core nodes	Constituent circuits	Functional role	Contested / notes
Default Mode (DMN)	mPFC, posterior cingulate/precuneus, angular gyrus/IPL, hippocampal formation, lateral temporal	MEM-01/03; overlaps REW-04 (vmPFC)	Self-referential thought, autobiographical memory, prospection, mentalizing, mind-wandering	"Task-negative" is oversimplified ; it's task-positive for internal tasks. Has ≥2–3 subsystems

Network	Core nodes	Constituent circuits	Functional role	Contested / notes
				(medial-temporal, dorsomedial-PFC, core).
				Boundary with the ventral-attention network is disputed; some treat SN as a "cingulo-opercular"/task-control network.
Salience (SN)	Anterior insula, dorsal ACC ($\pm$ VLPFC), amygdala, ventral striatum	INT-02, INT-01, THREAT-01, STR-02	Detects salient interoceptive/emotional/cognitive events; proposed to toggle DMN$\leftrightarrow$CEN	
Frontoparietal / Central Executive (CEN)	dIPFC, posterior parietal (IPS), dACC	EXEC-01/02/04, CSTC-ASSOC	Externally-directed control, working memory, decision-making, flexible task-sets	Overlaps the "multiple-demand" system; parcellations split it into control-A/B/C.
Dorsal Attention (DAN)	Frontal eye fields, intraparietal sulcus/superior parietal	DAN-01, CSTC-OCULO	Top-down/endogenous spatial attention, goal-driven orienting	Well defined.
Ventral Attention (VAN)	Temporoparietal junction, ventral frontal cortex (right-lateralised)	(overlaps INT-02)	Bottom-up reorienting to unexpected salient stimuli ("circuit-breaker")	Heavily overlaps SN; some frameworks merge them.
Sensorimotor (SMN)	Pre/postcentral gyri, SMA	CSTC-MOTOR, EXEC-03 (stop)	Somatomotor execution & somatosensation	Well defined; splits into face/hand/foot + a lateral network.

Network	Core nodes	Constituent circuits	Functional role	Contested / notes
Visual	Occipital (V1→ventral/dorsal streams)	(deferred — sensory pass)	Visual processing	Well defined.
Auditory	Superior temporal (A1→belt)	(deferred)	Auditory processing; abuts language	Well defined.
Limbic	OFC, temporal pole, ventral/medial temporal, amygdala	THREAT-01/04, REW-04, AFF-01, MEM-01	Emotion, valuation, memory, affiliation	Least reliable canonical network — low fMRI SNR (signal dropout in OFC/temporal pole); membership varies most across atlases.

Between-network dynamics — the triple-network model. The central dynamic claim is that network switching between the default-mode and central-executive networks, gated by the salience network, is required for tasks that alternate between internally- and externally-directed processing, with the anterior insula as the central node detecting salient stimuli and modulating the executive and default-mode networks. Under this model: DMN and CEN are broadly **anticorrelated**; the SN (anterior insula/dACC) detects a salient event and biases the system toward CEN engagement (external) or DMN (internal); aberrant switching is proposed to underlie symptoms across psychiatric disorders. [ScienceDirectResearchGate](#)

Where it's contested (state honestly): the *causal, directional* switching role of the anterior insula is supported by dynamic-causal-modelling and intracranial-EEG work but remains an **active area of research**, and recent methodological reviews of the intracranial evidence flag the interpretation as **not yet settled** — a large intracranial-EEG dataset used to support the salience node's directing role has been criticised as incomplete in methodology and interpretation. More broadly, the *number* of canonical networks is a choice of granularity (commonly 7 vs 17 in the Yeo/Schaefer parcellations), anticorrelation strength depends on preprocessing (global-signal regression), and network *boundaries* — especially SN/VAN and the limbic network — are genuinely disputed. [eLife](#)

4 · Developmental & plasticity layer

The parameterised, per-circuit schedule is in **developmental_parameters.csv** (27 rows: onset · milestone · functional_maturation · pruning_window · myelination ·

sensitive_period_gates · plasticity_type · plasticity_coeff · confidence · evidence_base). The rules a model would implement:

Rule 1 — Maturational sequence (subcortical → cortical, posterior → anterior).

Brainstem/arousal (AROU-*, THREAT-03) and limbic/subcortical circuits come online first (largely functional near birth); sensory and motor cortex mature in early childhood; association and prefrontal cortex last. Within the frontal lobe, maturation proceeds **back-to-front**, dlPFC/vmPFC (EXEC-01/02, THREAT-04) latest. Onset/growth parameters should therefore be ordered brainstem < limbic < sensory/motor < association < PFC.

Rule 2 — The "mid-20s" figure is a soft asymptote, not a switch. Prolonged PFC maturation is real, but the popular "brain matures at 25" claim is an oversimplification: the foundational longitudinal MRI work showed frontal grey matter peaks pre-adolescence and thins via pruning with the PFC among the last regions to mature, but that data stopped around age 20, so 25 was an estimated endpoint rather than a measured one, while white-matter and functional reorganisation continue for decades. Model this as a **growth curve asymptoting across the early-to-mid-20s with wide individual variance and no discontinuity**, not a step function. Human PFC dendritic-spine elimination in fact continues beyond adolescence into the third decade. [Wikipedia + 2](#)

Rule 3 — Synaptic overproduction then activity-dependent pruning, region by region. Each cortical area overshoots synapse number, then prunes back — but **on a region-specific schedule**: synaptic density in primary visual cortex peaks around 4–12 months, whereas prefrontal synaptogenesis peaks later (second-to-third year) and prunes through later childhood and adolescence, with thalamic regions completing earlier than neocortex, and sensory/language cortex maturing before prefrontal cortex which refines into early adulthood. Parameters: peak_synapse_age and pruning_window set **per circuit**, not globally. [Wiley Online LibraryPubMed Central](#)

Rule 4 — Pruning mechanism (with an honest caveat). Microglia phagocytose synapses; the classical complement (C1q/C3/C4) "eat-me" tag drives this in early development, and C4 over-dosage is a schizophrenia risk factor. **But** the popular story that complement drives *normal adolescent PFC pruning* is not well supported in the healthy human brain: human PFC complement-gene expression peaks between ~1–5 years and declines into adolescence, so complement may be redundant for normal adolescent PFC pruning and pathological chiefly under C4 over-expression. Implement complement-tagged pruning as **strong early-childhood, weak in normal adolescence**, activity-dependence dominating the adolescent window. [Frontiers](#)

Rule 5 — Hebbian activity-dependent wiring. "Fire together, wire together": co-active inputs strengthen (NMDA-dependent LTP), uncorrelated inputs weaken/prune (LTD). Two experience regimes to parameterise separately: **experience-expectant (EE)** plasticity waits for species-typical input during a critical window (e.g. patterned vision, basic language phonology, attachment) and *prunes if the input is absent*; **experience-dependent (ED)** plasticity encodes idiosyncratic experience across the lifespan (skills, episodic memories,

habits). Each circuit carries a `plasticity_type` $\in \{EE, ED, EE+ED\}$ and a `plasticity_coeff` that is high during any sensitive window and lower after.

Rule 6 — Critical/sensitive periods are opened and closed by inhibition. The molecular timer: critical-period onset is triggered by maturation of fast-spiking parvalbumin (PV+) GABAergic interneurons shifting the excitation/inhibition balance, and PV circuits mature at different times across regions, producing a sequential appearance of critical periods; closure is enforced by **molecular brakes** — perineuronal nets condensing around PV+ cells, whose experimental removal reinstates juvenile plasticity in adults, governed by opposing "trigger" and "brake" gene programs. So a model's per-circuit `sensitive_period_open/close` should be tied to that circuit's inhibitory-maturation timing, and windows should be **reopenable** in principle (a lever for intervention). What each gates (examples): ocular-dominance/binocular vision (early infancy, months), phoneme discrimination (first year), first-language syntax (early childhood, gradual close by puberty), attachment/threat-set-point and HPA programming (early life), amblyopia-relevant visual acuity (early childhood). [MDPI + 2](#)

Rule 7 — Adolescent limbic–prefrontal reorganisation. The reward/socioemotional system (REW-01/03, ventral striatum + vmPFC) matures **earlier** than the cognitive-control system (EXEC-01/02, lateral PFC/parietal/dACC), producing a transient window of heightened reward/sensation-seeking with still-maturing top-down control — the dual-systems/"maturational-imbalance" account: increased adolescent risk-taking is attributed to heightened reward sensitivity plus immature impulse control, tied to an early-maturing ventral-striatal/vmPFC socioemotional system and a late-maturing lateral-PFC/parietal/dACC control system. **Contested:** treat this as a strong heuristic, not settled law — critics note the imbalance hypothesis has limited falsifiability and weak direct links between the neural imbalance and real-world risk behaviour, arguing the models are informative heuristics rather than tightly testable theories. Parameterise as **two growth curves with an offset**, and flag the behavioural mapping as uncertain. [PubMed](#)
[CentralScienceDirect](#)

Rule 8 — Genetics × experience → divergence. Identical starting circuitry diverges because (a) common variants set *gains* on transmitter systems (e.g. dopamine-clearance, glucocorticoid-receptor, oxytocin-receptor tone) that scale each circuit's reactivity; (b) EE plasticity locks in whatever species-typical input arrives, and ED plasticity writes idiosyncratic history into synaptic weights; (c) *genexenvironment* interactions and epigenetic marks (e.g. early-life caregiving → HPA set-point via glucocorticoid-receptor methylation, in animal models) bias trajectories; (d) stochastic developmental noise and activity-dependent competition amplify small initial differences. In model terms: per-circuit **genetic priors** on reactivity/`plasticity_coeff`, times an **experience stream** integrated through Hebbian rules within each circuit's windows, yields individual weight matrices from a shared architecture.

Executive functioning — frontal-cortex pathways (Block T)

The high-level EXEC-01...04 circuits from Pass 1 are now decomposed into the actual prefrontal subregions, the white-matter tracts that wire them, the loops they run through, and the neuromodulation that tunes them.

PFC subregions and their executive functions

Node (ID)	Brodmann	Executive function	Key connectivity	Maps to
Dorsolateral PFC (PFC-DL)	9/46	Working-memory manipulation, planning, top-down goal control	↔ posterior parietal (via SLF); ↔ dorsal caudate (associative loop)	EXEC-01
Ventrolateral PFC (PFC-VL)	44/45/47	Controlled retrieval & selection; response inhibition (right)	↔ pre-SMA (via FAT); ↔ STN (hyperdirect)	EXEC-03 (part)
Dorsomedial PFC / dACC (PFC-DM)	8m/9m/24/32	Conflict & error monitoring, effort, expected value of control	↔ dlPFC, pre-SMA, LC	EXEC-02
Ventromedial PFC (PFC-VM)	10m/11m/14/25	Value integration, emotion regulation, self-reference	↔ amygdala/ant. temporal (via uncinate)	REW-04/THREAT-04
Orbitofrontal (PFC-OFC)	11/13/47	Outcome value, reversal learning, flexible behaviour	↔ ventromedial caudate (orbitofrontal loop); amygdala	~REW-04
Frontopolar / rostral PFC (PFC-FP)	10	Prospective memory, branching/multitasking, relational integration, metacognition; gateway between stimulus-oriented and stimulus-independent thought	↔ mid-lateral PFC; DMN	new

Node (ID)	Brodmann	Executive function	Key connectivity	Maps to
Frontal eye fields (FEF-01)	8	Oculomotor/spatial attention control	↔ IPS (dorsal attention); caudate (oculomotor loop)	DAN-01
Pre-SMA / SMA (PFC-preSMA)	6m	Action selection, sequencing, inhibition (stopping)	↔ IFG (via FAT); putamen (motor loop); STN	MOT-02/EXEC-03

Frontal white-matter tracts (the pathways themselves)

Tract (ID)	Connects	Function
Superior longitudinal fasciculus (TRACT-SLF, I/II/III) + arcuate	Frontal ↔ parietal (and ↔ temporal via arcuate)	Frontoparietal control, attention, language (dorsal stream)
Cingulum (TRACT-CING)	ACC/mPFC ↔ posterior cingulate / parietal / hippocampus	Cognitive control, default-mode integration, memory
Uncinate fasciculus (TRACT-UNC)	OFC/vmPFC ↔ anterior temporal / amygdala	Emotion–cognition integration, value, social-emotional; among the latest-maturing tracts (into the 30s)
Frontal aslant tract (TRACT-FAT)	Posterior IFG ↔ pre-SMA/SMA/ACC	Left: speech initiation, sequencing, verbal fluency (stimulation → speech arrest); right: inhibitory control
Frontostriatal fibres (TRACT-FS)	PFC ↔ striatum (anterior limb, internal capsule)	Carries the CSTC executive/limbic loops

Two organising principles folded in

- **Rostro-caudal control hierarchy.** Lateral PFC is organised along a posterior→anterior axis by *level of abstraction and temporal scale*: caudal (premotor) handles concrete stimulus–response control, mid-lateral handles contextual control, and rostral/frontopolar handles the most abstract, temporally-extended goals. Well supported by fMRI, lesion, and TMS work — with the live caveat that whether the rostral pole or mid-lateral PFC is the true *apex* is disputed.
- **Executive-function fractionation (process view).** The dominant individual-differences model is "unity and diversity": a **Common-EF factor** plus **Updating-**

specific and **Shifting-specific** components — with **inhibition showing no reliable variance beyond Common EF**. Mapping to circuits: *inhibition/Common-EF* → EXEC-02/03 + right vLPFC + pre-SMA→STN; *updating* → EXEC-01 + WM-02 (basal-ganglia gating); *shifting* → EXEC-01 + WM-02 + DAN-01; higher-order *planning/branching* → PFC-FP. Common EF maps onto the multiple-demand / frontoparietal system.

Neuromodulatory tuning of PFC (already-registered circuits, cross-referenced)

Dopamine (REW-02, mesocortical; D1 inverted-U — too little or too much impairs working memory), noradrenaline (AROU-01; α 2A improves signal-to-noise, but stress-level release impairs PFC), acetylcholine (AROU-03; attentional gain), serotonin (AROU-02; flexibility/impulse). This is why the frontal executive system is uniquely sensitive to stress and arousal state.

Model additions: PFC-FP added as a co-activation node; four executive states added (multistep_planning, set_shifting_task_switching, prospective_memory_branching, metacognitive_monitoring); frontopolar edges added to decision_under_uncertainty, working_memory Updating, theory_of_mind_mentalizing, rest_mind_wandering. Developmentally, frontopolar cortex and the uncinate tract are flagged as **the latest-maturing components of the entire model** (into the 20s–30s), which is the neural basis for late-maturing abstract reasoning, prospective planning, and metacognition.